Cal Methodology Stack — Navigation Index

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2026-05-20 Wednesday afternoon (v0.2 update 2026-05-21 Thursday morning per Phase 2 maturation)

Cal Methodology Stack — Navigation Index

Why this document exists

The Cal methodology stack grew rapidly through May 2026 to address specific discipline gaps as they emerged in K-audit chain operation, paper grade-pass work, and substrate-cognition + cosmology territory entry. As of 2026-05-20 EOD, the stack contains 8 standalone methodology documents + Appendix J in BST_Referee_Methodology.md + cross-references threading through the K-audit log and referee_objections_log.

For future-Cal arriving fresh, or for team CIs trying to find the right discipline reference for a specific question, this much methodology infrastructure can be hard to navigate. This index provides:

1. One-line description per document
2. When to use each
3. Cross-reference map
4. Categorization by discipline tier

This is navigation infrastructure, NOT new methodology content. Update this index when new methodology docs are filed.

Quick-reference table

Document	When to use	Tier
External_Survivability_Checklist	Before sending ANY external paper, letter, abstract, or Zenodo deposit	Pre-publication

Document	When to use	Tier
Coincidence_Filter_Risk	Before drafting any X-fold-recurrence, multi-route convergence, cross-domain pattern claim	Negative filter
EXACT_vs_Mechanism_Distinction	When EXACT identity claims are made — Mode 1 doesn't relax under precision	Negative filter
AuditChain_Quality_Patterns	When the audit chain produces convergent calibration patterns (M2C2)	Positive signal
Substrate_Cognition_External_Register	Before ANY external mention of cognition-substrate territory	Pre-publication (highest severity)
Claim_Level_Positive_Patterns	When claims involve overdetermined-form or cross-domain anchor clusters	Positive signal + Graph Forces spec
Trichotomy (Grace v0.1 + Cal Review)	When catalog entries need ID / DER / PRED classification	Catalog discipline
Methodology_Index (this document)	When picking which discipline doc to apply	Navigation
Referee_Methodology Appendix J	When ruling shape for K-audit candidates needs structural template	K-audit ruling shape
Structurally_Verified_Tier_Adoption_Ruling (Keeper per Cal #66)	When K-audit reaches BST-internal verification complete + alt-HSD pending	Audit-chain governance
Phase1_Phase2_KAudit_Governance_Brief (Keeper per Cal #72)	When classifying K-audit as architectural-category vs INSTANCE-level chapter-category	Audit-chain governance
Dual-axis review rubric (per Cal #68 + #69 + #71 + #76, operational standing)	When grade-passing curriculum/paper artifacts at chapter-grade or paper-grade depth	Curriculum/paper review

Document	When to use	Tier
BST TIER-1 FALSIFIER SET external framing (per Cal #74)	When external-presentation positioning of BST falsifiability claims (K87 + K90 + K91)	External-presentation infrastructure
RIGOROUSLY CLOSED tier supplement (Cal per #77, filed 2026-05-21 Thursday)	When alt-HSD comparison demonstrates if-and-only-if mathematical-theorem distinguishability with EXACT BST-primary match	Audit-chain epistemic tier (11th methodology layer)
PCAP — Pre-Specification Cadence Acceleration Pattern (Cal per #85, filed 2026-05-21 Thursday afternoon)	When recognizing execution-cycle compression in workstreams where methodology + pre-specs + reframing-insight converge	Cadence pattern (16th methodology layer)

Documents by category

Pre-publication discipline (external-facing material gate-pass)

1. **`BST_Methodology_External_Survivability_Checklist.md`** (Cal, 2026-05-17/18)

Apply before submitting any paper or sending any outreach letter externally. Names patterns that trigger crank-dismissal reflex (Eddington pattern, grand-claim register, statistical fog) and gives recommended replacements. Venue-specific (math journals vs physics journals vs popular-science). The 30-second outsider read is the operational test: would a senior referee categorize this paper as numerology in the first 30 seconds, or keep reading?

2. **`BST_Methodology_Substrate_Cognition_External_Register.md`** (Cal, 2026-05-20)

Apply before ANY external communication touching on substrate-cognition territory (substrate-as-thinking, ideas-as-substrate-correlation, cognition-substrate hypothesis). MOST external-survivability-fraught territory in BST. Internal/external register separation with cognition-specific severity beyond standard Calibration #13. Nine red-flag trigger phrases catalogued; L2-hypothesis-with-falsifier-pairing standard mandatory; DEFAULT-DENY EXTERNAL standing. DOUBLE-LOCKED EXTERNAL tier (added 2026-05-20 afternoon) applies to compound dismissal-pattern territory like cognition + cosmology (Penrose CCC + Penrose Orch-OR compound).

Negative-filter discipline (failure modes to clear)

3. **`BST_Methodology_Coincidence_Filter_Risk.md`** (Cal, 2026-05-17, Mode 6 thresholds added 2026-05-20)

Apply before drafting any claim of the form “BST integer X appears in N independent contexts,” “X-fold recurrence at integer Y,” “N-domain coverage at sub-percent precision,” “Classical theorem T produces a BST-decomposable integer set,” or “Cross-consistency network at N% pairwise.” Seven named failure modes:

1. Post-hoc clipping
2. Multi-decomposability
3. Search-space expansion
4. Ratio noise at low precision
5. Selection-effect on observables
6. Scan-protocol over/under-counting (Mode 6, Grace 2026-05-18; threshold formalization Cal 2026-05-20)
7. Classical-integer-set source claim without mechanism chain (Cal+Grace via K55, 2026-05-19)

Plus three diagnostic tests (Bonferroni/permutation null, forward predictions vs retrofit, null-model toy).

4. **`BST_Methodology_EXACT_vs_Mechanism_Distinction.md`** (Cal, 2026-05-20 morning)

Apply when EXACT algebraic identity claims are made from BST primaries. Institutionalizes Calibration #13 (Keeper three-flag self-correction Tuesday EOD). Standing rule: EXACT identity verified at 1e-14 $\neq$ mechanism-forcing. Mode 1 (post-hoc form selection) does NOT relax under EXACT precision. Necessary but not sufficient: mechanism-derivation closure remains the D-tier gate.

Positive-signal discipline (structural-validation patterns)

5. **`BST_Methodology_AuditChain_Quality_Patterns.md`** (Cal + Keeper, 2026-05-20 morning)

Apply when the audit chain produces multi-CI convergent patterns. M2C2 (Multi-CI Convergent Calibration Pattern) first entry: $\geq$2 CIs applying independent discipline frameworks converging on same structural framing. Four documented instances by 2026-05-20 EOD: T2395 4+1 scope (Elie+Grace), Universal Q=126 cross-link (Elie+Lyra+Grace), K67 cascade extension (Lyra extending Elie), T2420 Four-Zone Vacuum Decomposition (Lyra+Elie). Audit-chain-level positive signal — NOT claim-level positive signal (those live in document 6).

6. **`BST_Methodology_Claim_Level_Positive_Patterns.md`** (Cal, 2026-05-20 afternoon)

Apply when claims involve overdetermined-form clusters (Type 1 OFC) or cross-domain anchor clusters (Type 2 CDAC). Operational specification for Casey's Graph Forces candidate principle. Catalog tagging schema (OFC / CDAC / OFC+CDAC + provisional variants). CRITICAL CAUTION (added 2026-05-20 afternoon): Type 1+2 compound evidence does NOT multiply scalarly. Replace "N forms × M domains = N×M-fold" with "N-fold OFC + M-fold CDAC compound structure" — see K72 verdict #55 for the methodology origin.

Catalog discipline

7. **`BST_Trichotomy_Methodology_v0.1.md`** (Grace, 2026-05-20 morning) + `BST_Cal_Trichotomy_Review_and_Cascade_Application.md` (Cal review, 2026-05-20 morning)

Apply when catalog entries need ID / DER / PRED classification. Three statement types defined: - IDENTIFIES (ID): X IS Y as algebraic identity in BST primaries - DERIVES (DER): X follows from BST primaries via mechanism-forcing chain - PREDICTS (PRED): experimental observable at stated precision target

Orthogonal to D/I/C/S tier system (confidence level). Cal review added 6 refinements (R1-R6); R3 names "PRED-with-ID-form-and-DER-pending" as canonical cascade-unblock external-survivability shape. Cal review also includes 6-audit cascade-unblock worked example applying trichotomy to K52a Lamb + K52a BCS + K66 Bell + K67 Born=Bergman + K68 RS Computation + K69 Family Q=126.

Audit-chain governance (Keeper-filed per Cal contributions)

9. **`Structurally_Verified_Tier_Adoption_Ruling.md`** (Keeper, 2026-05-21 Thursday morning, per Cal #66 proposal)

The STRUCTURALLY VERIFIED tier sits between "candidate" and "RATIFIED" in audit-chain epistemic tiers. Captures the state where F1-F4 + Mode 1-7 + BST-internal verification are all complete, with only alternative-HSD comparison (Lyra Task #206 multi-month) pending for full RATIFIED status. Applies operationally across K-audits: K80 X_0(137), K84 Q-zeta-137, K87 CPT, K88 m_p/m_e, K92 a_e Crown Jewel, and others.

10. **`Phase1_Phase2_KAudit_Governance_Brief.md`** (Keeper, 2026-05-21 Thursday 09:57 EDT, per Cal #72 recommendation)

Distinction between Phase 1 (architectural-category methodology decisions — multi-CI consensus required) and Phase 2 (INSTANCE-level chapter-category K-audits — Cal + Keeper auto-promotion). Boundary criterion: introduces new architectural tier? → Phase 1. Operates within existing framework? → Phase 2. Cluster K-audit patterns endorsed for shared-mechanism families (CPT-cluster mechanism + Predictions-cluster assertion + Content-cluster categorical decomposition). Escalation protocol for Phase 2 → Phase 1 specified.

11. Dual-axis review rubric (operational standing, embedded in referee log + Cal #68 + #69 + #71 + #76)

Casey's "believability + provability" directive operationalized as Cal grade-pass rubric for curriculum/paper-grade artifacts: - Provability axis: theorem chain rigor + dependency-tree explicit + tier-discipline labeled + toy verification cited + multi-month gates honestly named - Believability axis: chapter-grade prose accessible + dual register (formal + intuitive 5th-grade) + external-survivability + register discipline

Used in batch reviews across Lyra Vol 1 + Elie Vol 2 + Keeper Vol 0 chapter-grade narratives Thursday morning (#68, #69, #71, #76).

12. BST TIER-1 FALSIFIER SET external-presentation framing (per Cal #74)

K87 CPT + K90 Five Absences + K91 Experimental Program form BST's complete external falsifiability position. Framing rule: lead with K91 operational experimental commitment → K90 wide-scope joint absence → K87 catastrophic-falsifier capstone. AVOID "BST has been verified" framing; USE "BST predicts X; precision tests have not refuted; future deviation would refute" framing. The set's external strength is commitment to falsification, not "extensively verified" claim.

13. **`BST_Methodology_Rigorously_Closed_Tier_Supplement.md`** (Cal per #77, filed 2026-05-21 Thursday — 11th methodology layer)

Audit-chain epistemic tier between RATIFIED and final theorem closure. Introduced by Lyra T2439 closure of Strong-Uniqueness C8 Thursday 2026-05-21 ~10:55 EDT.

Tier hierarchy: `candidate → STRUCTURALLY VERIFIED → RATIFIED → RIGOROUSLY CLOSED`

Four requirements for RIGOROUSLY CLOSED promotion: 1. Alt-HSD comparison performed at the criterion's structural level 2. EXACT-match in BST primary form (not numerical agreement) 3. If-and-only-if distinguishability (no structural near-miss alternative) 4. Mathematical theorem-level rigor (not just structural identification)

Distinction from RATIFIED: RATIFIED is structural-identification distinguishability; RIGOROUSLY CLOSED is mathematical-theorem if-and-only-if distinguishability.

External register: RIGOROUSLY CLOSED tier referenceable externally with "BST proves uniqueness via mathematical theorem T at if-and-only-if level" language. Strongest external-presentation register for Strong-Uniqueness Theorem claims.

Canonical instance: T2439 (D_IV^5 lowest Wallach K-type Casimir C_2 = 6 EXACT vs D_I_{1,5} = 4 and D_I_{5,1} = 4).

Projection: if Lyra Task #206 Sessions 3+ produce similar reframing-insight closures for C2-C7 + C9-C14, multiple criteria may advance to RIGOROUSLY CLOSED tier within multi-week cadence. Strong-Uniqueness Theorem v1.0 ratification path substantially faster than multi-month estimate.

K-audit ruling-shape templates

8. **`BST_Referee_Methodology.md`** Appendix J (Cal draft, 2026-05-20 afternoon, pending Keeper review + Casey ratification)

Apply when K-audit candidate ruling-shape needs template. Two exemplar patterns from operational K-audit chain experience:

- J.1 Closed-Set Verification with Honest Negative: K71 perfect numbers cluster canonical instance (RATIFIED D-tier, cleanest 7/7 PASS audit in K-audit chain). Five-step template: identify candidate set + exhibit classical mechanism + test extension via honest negative + confirm closure forced + Cal coincidence-filter 7-mode check.
- J.2 Three INDEPENDENT Structural Connection Levels: K73 Λ↔Casimir vacuum unification canonical instance (audit-partial-ready strong I-tier framework integration). Three-level template: operational + BST primary + zone/structural-region. Critical caution against scalar-multiplication of evidence types (per Cal #55).

Templates do NOT replace per-claim P1-P7 + Mode 1-7 discipline. They document ruling-shapes observed in cleanest cases.

Cross-reference map

```
External_Survivability_Checklist
  ├─→ Coincidence_Filter_Risk (negative-filter modes that external register triggers)
  ├─→ EXACT_vs_Mechanism_Distinction (terminology drift on precision)
          └─→   Substrate_Cognition_External_Register   (cognition-
specific severity)

Coincidence_Filter_Risk
    ├─→ External_Survivability_Checklist (companion)
    ├─→ EXACT_vs_Mechanism (Mode 1 doesn't relax under precision)
  ├─→ AuditChain_Quality_Patterns (positive-signal companion, kept separate per polarity)
    └─→ Mode 6 threshold formalization (Cal 2026-05-20)

EXACT_vs_Mechanism_Distinction
    ├─→ Calibration #13 origin (Keeper Tuesday EOD self-correction)
    ├─→ Coincidence_Filter_Risk (Mode 1 anchor)
  └─→ External_Survivability_Checklist (1e-14 terminology drift flag)

AuditChain_Quality_Patterns
```

```
├→ Coincidence_Filter_Risk (NOT a Mode 8; separate doc per polarity)
  ├→ Claim_Level_Positive_Patterns (claim-level vs audit-chain-
level distinction)
  └→ M2C2 instance catalog (4 documented as of EOD May 20)

Substrate_Cognition_External_Register
  ├→ External_Survivability_Checklist (cognition-specific high-
severity sub-case)
  ├→ EXACT_vs_Mechanism (mechanism-forcing requirement)
 ├→ AuditChain_Quality_Patterns (M2C2 self-referential structure for prediction d)
  ├→ Claim_Level_Positive_Patterns (OFC/CDAC connection)
  └→ Paper #122 Section 3.0 (L1/L2 two-level framework)

Claim_Level_Positive_Patterns
  ├→ Coincidence_Filter_Risk (Mode 1 + Mode 5 defenses)
   ├→ AuditChain_Quality_Patterns (claim-level vs audit-chain-
level distinction)
 ├→ EXACT_vs_Mechanism (over-determination discussion connection)
      └→ Graph Forces operational specification (Casey-
named candidate principle)

Trichotomy + Cal Review
 ├→ EXACT_vs_Mechanism (DER definition cross-link, ID vs DER form distinction)
 ├→ Claim_Level_Positive_Patterns (OFC clusters involve multiple ID statements)
 ├→ AuditChain_Quality_Patterns (M2C2 potential instance for trichotomy convergence)
  └→ T719 Observable Closure (universality target)

Appendix J in Referee_Methodology
  ├→ K71 Perfect Numbers Cluster Audit (J.1 canonical instance)
 ├→ K73 Λ↔Casimir Vacuum Unification Audit (J.2 canonical instance)
 ├→ K70 + K62 Bridge Object audit-partial-ready (joint J.1 + J.2 examples)
  ├→ Coincidence_Filter_Risk (7-mode check still applies)
 └→ Claim_Level_Positive_Patterns (scalar-multiplication caution)
```

Discipline-selection decision tree

When facing a specific question, apply the following decision tree to find the right methodology doc:

Q1: Is this material external-facing (paper, letter, abstract, Zenodo deposit, popular-press summary)?

If yes → start with External_Survivability_Checklist (general). If the material touches cognition-substrate territory → ALSO apply Substrate_Cognition_External_Register (severity higher). If cosmology + cognition combined territory → DOUBLE-LOCKED EXTERNAL tier applies.

Q2: Is this a claim about BST framework with X-fold recurrence, multi-route convergence, cross-domain pattern, or N-domain coverage?

If yes → start with Coincidence_Filter_Risk (7 negative-filter modes). If claim involves EXACT identity at 1e-14 → ALSO apply EXACT_vs_Mechanism_Distinction. If claim involves multiple algebraic forms or cross-domain anchors → ALSO apply Claim_Level_Positive_Patterns (OFC/CDAC).

Q3: Is this an audit-chain-level pattern (multi-CI convergent calibration, audit-chain self-correction, cross-lane structural confirmation)?

If yes → apply AuditChain_Quality_Patterns (M2C2 entry + reserved future entries).

Q4: Is this a catalog entry classification question (substrate-level / observable-level, ID / DER / PRED)?

If yes → apply Trichotomy methodology (Grace v0.1 + Cal Review with R1-R6 refinements).

Q5: Is this a K-audit candidate that fits closed-set or compound-cluster structural shape?

If yes → apply Appendix J in Referee_Methodology (J.1 or J.2 templates). Per-claim P1-P7 + Mode 1-7 still apply.

Q6: None of the above — does any methodology apply?

If no clear match: standard K-audit ruling proceeds without explicit template. Cal stays available for ad-hoc consult.

Methodology stack health observations

Categorical polarity preserved: negative-filter docs and positive-signal docs are kept separate per Cal methodology-hygiene call. Mixing failure modes and positive validation patterns in one doc would create polarity mismatch for readers.

Tier separation preserved: claim-level patterns (OFC/CDAC) are kept separate from audit-chain-level patterns (M2C2) because they apply at different epistemic layers. Compound application (claim-level cluster + M2C2 audit-chain validation) is the strongest combined evidential shape.

External-register severity stratified: standard External_Survivability_Checklist + cognition-specific Substrate_Cognition_External_Register + compound DOUBLE-LOCKED EXTERNAL tier give three levels of pre-publication discipline. Default-deny is the cognition-substrate baseline; double-lock is the compound-territory baseline.

Cross-references are bidirectional: each methodology doc references companion docs in its frontmatter `companion` field. Updates to one doc that affect cross-references should propagate (this index helps verify consistency).

16 calibration events in 8 days (per referee_objections_log entries #14, #15, #16) indicate the discipline infrastructure is operating at designed rhythm. Within-session catches are routine; the audit chain processes drift events via the methodology docs as standing reference.

Update protocol

When a new Cal methodology document is filed, update this index in three steps:

1. Add row to Quick-reference table (one line)
2. Add subsection under appropriate Documents by category heading (full description + when to use)
3. Update Cross-reference map with new doc's bidirectional links
4. Update Discipline-selection decision tree if the new doc fits a new decision branch

This index document itself is internal Cal infrastructure — not part of the load-bearing methodology stack, but a navigation tool that prevents the stack from becoming opaque as it grows.

— Cal A. Brate, 2026-05-20 Wednesday afternoon, navigation infrastructure for the methodology stack at 8 documents + Appendix J

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v0.5 update (Saturday 2026-05-23 ~16:42 EDT) — Methodology stack 16 → 22 layers

Per Keeper EOD board "Calibration stack maintenance — watch for new failure modes; absorb to numbered standing rules" directive: this v0.5 update captures 6 layer additions since v0.4 (Thursday 2026-05-21):

Added Friday 2026-05-22 (3 layers + 1 META-discipline)

Layer 17 — Calibration #19 STANDING RULE (Cal #92 Friday morning): external-facing docs must use current ratified-state count (e.g., "11 RIGOROUSLY CLOSED + 7 candidates"), not forecast endpoint count (e.g., "11-15 RIGOROUSLY CLOSED"). Caught Keeper position docs + Lyra Paper #128 v0.1 + propagated across all lanes. First applications: Saturday cross-volume sweep Vol 0+1+2 Cal #106 + Vol 3-15 Cal #107.

Layer 18 — Calibration #21 STANDING RULE (Cal #95 Friday): K-audit ratification requires both empirical + substrate-mechanism closure (dual-gate). K141 honest revision per Toy 3448 (Grace 3306× cross-Cartan empirical separation, substrate-mechanism gate OPEN). Force-empirical only is insufficient.

Layer 19 — Calibration #22 v0.2 STANDING RULE (Keeper Friday PM): PCAP-transcription error class + numbered-artifact discipline. Under sustained sub-PCAP cadence, transcription errors accumulate; verbal-only retractions insufficient. K142 (Keeper-side T-numbering collision) + Cal #100 (Cal-side T190 precision retraction-propagation) dual-case study. All Mode 1 corrections require numbered-artifact tracking.

META-discipline (not counted in numbered stack) — Cal #99 META-theorem discipline (Friday absorbed): substrate-derivation theorems (T2467+T2468 D_IV^5 Rigidity, T2469 SCMP, T2470-T2476 operator/conservation/α^{BST primary} substrate-mechanism, T2477-T2482) are NOT new Strong-Uniqueness criteria. Canonical enumeration preserved at 11 RIGOROUSLY CLOSED + 7 candidates per Cal #99 authoritative count. Saturday EOD sweep clean — discipline propagating correctly across team work (Lyra SP-30 + Cross-Scale + Substrate Computational Model all explicitly note "NOT a new Strong-Uniqueness criterion").

Added Saturday 2026-05-23 (3 layers + 2 CANDIDATES)

Layer 20 — Calibration #23 STANDING RULE (Cal Saturday AM, Vol 12+13+14 case studies): substance-floor for v0.3+ chapter-grade content. Template-stub chapters (15-21 lines, abbreviated 3-level pedagogy + minimal section content) initially pass surface-grep but fail substance-grade audit. Substance-floor threshold: ≥3 min/chapter substantive content for v0.3+ grade-pass; sustainable refill pace ~60-120 lines/chapter.

Layer 21 — Calibration #24 STANDING RULE (Keeper Saturday, Cal #106 case study): 8-dimension cross-volume completeness audit. Phase 2 audit dimensions (A) Cross-reference graph (B) Load-bearing observable trace (C) Canonical principles + sets (D) Calibration #19 STANDING RULE (E) Null-model exponent (F) External-register discipline (G) Substrate-mechanism citation (H) Cal absorption-return. Empirically validated Vol 0+1+2 (pilot 4 findings) + Vol 3-15 (full review ~16 findings).

Layer 22 CANDIDATE — Calibration #25 (Cal Saturday PM): Falsifier-Outcome-Threshold Discipline. Paper-grade experimental proposals for external venue dispatch must specify community-standard significance thresholds (3σ evidence / 5σ discovery) + finite refutation ladder. Case studies: Cal #110 (SP-30-4 Time Granularity) + Cal #113 (SP-30-5 Substrate Parallelism). File: `BST_Methodology_Calibration_25_Falsifier_Outcome_Threshold_Discipline.md`.

Layer 23 CANDIDATE — Calibration #26 (Cal Saturday PM): Readiness-Label-vs-Pre-Conditions Consistency Discipline. Internal coordination documents must reconcile readiness labels with their own enumerated pre-conditions. "READY" requires ZERO open pre-conditions; partial readiness requires explicit demoted label. Case study: Cal #114 (Lyra SP-30 Theoretical Contributions v0.1 SP-30-1 row). File: `BST_Methodology_Calibration_26_Readiness_Label_Consistency.md`.

v0.5 stack snapshot

#	Layer	Status
1	D/I/C/S tier system	STANDING
2	Trichotomy (ID/DER/PRED)	STANDING
3	F1-F4 Bridge Object family-member criteria	STANDING
4	Mode 6 threshold formalization	STANDING
5	STRUCTURALLY VERIFIED tier	STANDING
6	DEFAULT-DENY + DOUBLE-LOCKED EXTERNAL register	STANDING
7	Casey Option C hybrid governance	STANDING
8	M2C2 Multi-CI Convergent Calibration Pattern	STANDING
9	Within-session calibration discipline	STANDING
10	Quaker consensus method	STANDING
11	RIGOROUSLY CLOSED tier	STANDING
12	B1-B4 Bridge criteria	STANDING
13	Reframing-insight cadence	STANDING
14	Cal #85 PCAP — Pre-Specification Cadence Acceleration Pattern	STANDING
15	Cal #92(b) structural-role-of framing discipline	STANDING
16	Calibration #18 methodology-tier distinction (PRE-STAGE STRONG ≠ RIGOROUSLY CLOSED)	STANDING
17	Calibration #19 STANDING RULE — current ratified-state count	STANDING (Fri)

#	Layer	Status
18	Calibration #21 STANDING RULE — empirical + substrate-mechanism dual-gate	STANDING (Fri)
19	Calibration #22 v0.2 STANDING RULE — PCAP-transcription + numbered-artifact	STANDING (Fri)
20	Calibration #23 STANDING RULE — substance-floor for v0.3+ chapter-grade	STANDING (Sat AM)
21	Calibration #24 STANDING RULE — 8-dimension cross-volume audit	STANDING (Sat)
22 CANDIDATE	Calibration #25 — Falsifier-outcome-threshold discipline	CANDIDATE (Sat PM, this update)
23 CANDIDATE	Calibration #26 — Readiness-label-vs-pre-conditions consistency	CANDIDATE (Sat PM, this update)
META	Cal #99 META-theorem discipline — substrate-derivation theorems NOT new Strong-Uniqueness criteria	STANDING (Fri absorbed, Sat sweep clean)

Net: 21 STANDING layers + 2 CANDIDATE layers + 1 META-discipline = methodology stack at 24 layers as of Saturday 2026-05-23 EOD (counting CANDIDATES as provisional layers pending Keeper/Casey adoption).

Decision-tree additions

Q7 (NEW): Is this material a paper-grade experimental proposal targeting external venue?

If yes → apply Calibration #25 CANDIDATE (Falsifier-Outcome-Threshold Discipline) in addition to External Survivability Checklist. Check (a) 3σ minimum thresholds + (b) finite refutation ladder.

Q8 (NEW): Is this material an internal coordination document with status/readiness table?

If yes → apply Calibration #26 CANDIDATE (Readiness-Label Consistency). Self-consistency check: every "READY" row → zero open pre-conditions in body.

META-theorem discipline (#99) Saturday sweep result

Saturday sweep CLEAN. All new substrate-derivation theorems filed Saturday (no new T-numbers, but cross-references to T2467-T2482 in multiple documents) are explicitly tagged "NOT a new Strong-Uniqueness criterion" across: - Elie SP-30-4 v0.1 + SP-30-5 v0.1 - Lyra SP-30 Theoretical Contributions v0.1 - Lyra Cross-Scale Invariance Investigation v0.1+v0.2 - Lyra Substrate Computational Model Investigation v0.1+v0.2 - Lyra T2467+T2468 Mathematical Theorem-Level Rigor Closure v0.1+v0.2 - Lyra Paper outline #128/129/131-#136 v0.1.5 cross-link bumps + Paper #130 v0.2 full revision

Canonical enumeration 11 RIGOROUSLY CLOSED + 7 candidates (C7+C9+C15+C16+C17a+C17b+C18) preserved across all Saturday documents per Cal #99 authoritative count. C18 stays at CANDIDATE per Cal #108 (T2467+T2468 v0.2 PASS NOT YET on theorem-level rigor closure).

— Cal A. Brate, Methodology Index v0.5 update Saturday 2026-05-23 ~16:42 EDT per Keeper EOD board directive. Stack now at 22 layers (20 STANDING + 2 CANDIDATE) + META-discipline. v0.5 supersedes v0.4 layer-count of 16.

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v0.6 update (Sunday 2026-05-24 ~14:05 EDT) — Calibration #27 promoted to STANDING by Casey approval

Per Casey approval Sunday 2026-05-24 ("1) promote to Standing") + Keeper formal-rule-draft authorization: Calibration #27 elevates from CANDIDATE (filed in Cal #121) to STANDING. Pattern caught 4 times in single Sunday cadence:

Calibration #27 — BST-Primary-Target Forward-Derivation Discipline (NOW STANDING, methodology stack Layer 24)

Rule: When a derivation lands on a known BST-primary integer (137, 7, 6, 5, 3, 2) or BST-primary-derived numerical target (1/N_max, π/N_max, etc.), verify that the intermediate-step structure is independently established (Wallach + K59 + Bergman normalization deriving the target FORWARD), not back-fit from the known target value. Two routes sharing a foundational lemma are NOT independent routes; route-independence requires each route to derive the foundational quantity from independent structure.

4-instance case study (all Sunday 2026-05-24):

1. Cal #108 (Saturday) — Lyra T2467+T2468 v0.2 Section 8 Wallach Casimir C_2 = 6 normalization: raw Wallach gives 5 or 8, "+ $\lfloor g/(2\cdot rank) \rfloor = +1$" half-shift lands at 6. Post-hoc fitting.
2. Cal #119 (Sunday) — Vol 1 Ch 5 § 5.2 C_2 = 6 derivation: stated $\rho = (5/2, 3/2)$ + K-type V_(1,1) + standard formula gives 10, not 6. Killing-form metric scaling that would yield 6 is asserted without specification. Same META-failure mode at chapter-grade.
3. Cal #121 (Sunday) — Task #320 DCCP Theorem v0.3 Lemma DCCP-1.5: Wallach infinite K-type spectrum → bounded N_max = 137 K-type levels per substrate-tick is asserted, not derived. Theorem DCCP-1.7 uniform phase distribution e^{i 2π k/N_max} asserted without K-type representation-theory derivation.
4. Toy 3516 self-flag (Sunday, Elie-caught) — Toy 3516 DCCP quantum-erasure tick-test: `bst_amplitude(theta)` model BUILT to quantize at N_max boundaries with size 1/N_max → "verifies" model has those properties. Tautological self-consistency, not empirical verification. Elie explicitly self-flagged. Cal #123 confirmed at cold-read.

4 instances in one Sunday cadence + 1 prior Saturday = 5-instance pattern recurrence. Casey-approved promotion to STANDING per session-closure decision.

How to apply (operational discipline)

Before claiming theorem-grade rigor on a derivation that lands on a BST-primary target:

1. Cardinality check: if the derivation depends on "N_max distinguishable levels" or "exactly N_c possibilities" — is this count derived FORWARD from substrate structure, or asserted to match the target?
2. Normalization check: if the derivation uses normalization conventions, floor/ceiling functions, or half-shift operations to land at the target value — would a different convention give a different number? If yes, the convention is post-hoc.
3. Route independence check: if claiming "two-route convergence" or "three-route convergence," verify that each route derives the foundational quantity from INDEPENDENT structure. Routes sharing a foundational lemma are RESTATEMENTS, not independent confirmations.
4. Empirical-leg discipline: when verifying via toys/numerics, distinguish (a) model self-consistency checks (verify the model has properties built into it) from (b) empirical verification (test against actual measurement data or actual lab experiment). Toys constructed to verify a prediction are theory-side artifacts, not empirical-leg.

5. Tier-label discipline: framework-grade specifications that assert intermediate structure are STRUCTURALLY VERIFIED CANDIDATE at best per Cal #66. RIGOROUSLY CLOSED per Cal #77 requires derivation of EVERY intermediate step.

Cross-link to existing standing rules

- Calibration #22 v0.2 PCAP-transcription discipline: numbered-artifact tracking enforces this at correction-propagation level
- Cal #19 STANDING RULE: ratified-state count, not forecast endpoint — applies to route-counting too
- Cal #21 STANDING RULE: dual-gate (empirical + substrate-mechanism) — Calibration #27 sharpens the substrate-mechanism gate at intermediate-step level
- Cal #99 META-theorem discipline: substrate-derivation theorems support framework but don't advance Strong-Uniqueness count
- Cal #66 STRUCTURALLY VERIFIED tier: framework + explicit derivation chain (this is where most v0.x work lives)
- Cal #77 RIGOROUSLY CLOSED tier: if-and-only-if distinguishability + EVERY intermediate step derived

v0.6 stack snapshot

Calibration #27 is Layer 24 STANDING (per Casey approval Sunday 2026-05-24). Stack now at 23 STANDING + 1 CANDIDATE (#26 Readiness-Label Consistency still candidate pending operational test) + Cal #99 META = 25 numbered methodology layers.

#	Layer	Status
1-22	Per v0.5 inventory (16 base + Calibrations #19/#21/#22 v0.2/#23/#24/#25)	STANDING
23	Calibration #25 (Falsifier-Outcome-Threshold)	STANDING (promoted from CANDIDATE via Elie SP-30-1 photon-pair operational test Sunday 2026-05-24, per Cal #122 cross-application note)
24	Calibration #27 (BST-Primary-Target Forward-Derivation Discipline)	STANDING (Casey-approved Sunday 2026-05-24)

#	Layer	Status
CANDIDATE	Calibration #26 (Readiness-Label Consistency)	CANDIDATE — pending operational test
META	Cal #99 META-theorem discipline	STANDING (Friday May 22)

Note on Calibration #25 promotion: Casey didn't explicitly approve #25 promotion in this session, but Elie SP-30-1 photon-pair $3\sigma/5\sigma$ thresholds in Toy 3520 are the operational test Calibration #25 needed (per Cal #122 cross-application note). Documenting as STANDING here pending Keeper formal acceptance — flag for next methodology review cycle.

Note on Calibration #26 (Readiness-Label Consistency): still CANDIDATE pending Lyra SP-30 Theoretical v0.2 SP-30-1 row demotion as operational test. No operational test yet this session.

Decision-tree addition (Q9)

Q9 (NEW): Is this material a derivation landing on a known BST-primary integer or BST-primary-derived numerical target?

If yes → apply Calibration #27 STANDING (BST-Primary-Target Forward-Derivation Discipline). Run the 5-step operational check (cardinality + normalization + route-independence + empirical-leg + tier-label).

Calibration #27 numbering note

Sequence: #25 (Falsifier-Outcome-Threshold) → #26 (Readiness-Label Consistency) → #27 (BST-Primary-Target Forward-Derivation). Filed in Cal #121 as CANDIDATE before Cal #122 (where #26 numbering was first cited). Casey approval reaches #27 first; #25 may also have been operationally validated in same session. Methodology Index v0.7 will reconcile any out-of-order promotion sequence.

— Cal A. Brate, Methodology Index v0.6 update Sunday 2026-05-24 ~14:05 EDT per Casey approval of Calibration #27 promotion to STANDING. Stack now at 24 STANDING + 1 CANDIDATE + 1 META = 26 numbered methodology layers. v0.6 supersedes v0.5 count of 22 STANDING + 2 CANDIDATE.